VII

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

23BCE Reg. No 1463

Final Assessment Test(FAT) - NOV/DEC 2025

Programme	**B.Tech.**	Semester	**Fall Semester 2025-26**
Course Code	**BMAT205L**	Faculty Name	**Prof. Vidhya V**
Course Title	**Discrete Mathematics and Graph Theory**	Slot	**C2+TC2+TCC2**
		Class Nbr	**CH2025260100894**
Time	**3 hours**	Max. Marks	**100**

Instructions To Candidates

- Write only your registration number in the designated box on the question paper. Writing anything elsewhere on the question paper will be considered a violation.

Course Outcomes

CO1: Learn proof techniques and concepts of inference theory
CO2: Use algebraic structures in applications
CO3: Counting techniques in engineering problems.
CO4: Use lattice and Boolean algebra properties in Digital circuits.
CO5: Solve Science and Engineering problems using Graph theory.

Answer any 10 Questions (10 × 10 Marks)

01. a) Show that the premises ``A horse that is registered for today's race is not a thoroughbred" and ``Every horse registered for today's race has won a race this year" imply the conclusion ``Therefore a horse that has won a race this year is not a thoroughbred". (7 marks)
b) Write the following English sentence symbolically "Every city has at least one hospital" (3 marks)
[10] (CO1/K4)

02. a) Determine whether the following proposition is a tautology, contradiction, or contingency without using truth table.$\neg(p \vee (q \wedge r)) \leftrightarrow ((p \rightarrow q) \wedge (p \vee r))$. (5 marks)
b) How many 6-digit ATM passwords can be formed using the digits 0 to 9 if at least two digits are the same? (5 marks)
[10] (CO3/K2)

03. a) Prove that the set

$$H = \left\{ A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \middle| \, a, b, c, d \in \mathbb{R}, Det\ A = 1 \right\}$$

is a subgroup of the group $GL(2, \mathbb{R})$ under matrix multiplication, where

$$GL(2, \mathbb{R}) = \left\{ A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \middle| \, a, b, c, d \in \mathbb{R}, Det\ A \neq 0 \right\}$$

. {7 marks}
b) Let $\mathbb{R}$ denote the set of real numbers and $GL(2, \mathbb{R})$ denote the set of all non-singular matrices (invertible matrices). Let $f : (\mathbb{R}, +) \rightarrow GL(2, \mathbb{R})$ be the function defined by $f(x) = \left\{ \begin{bmatrix} e^x & 0 \\ 0 & e^{-x} \end{bmatrix} : x \in \mathbb{R} \right\}$. Prove that f is a group homomorphism.(3 marks)
[10] (CO2/K4)

04. Given the generator matrix, $G = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$ corresponding to the encoding function, $e : B^3 \rightarrow B^6$.
Find the corresponding parity check matrix and use it to decode the following received words and hence, to find the original message. Are all words decoded uniquely? (i)110101 (ii)111111 (iii)001111.

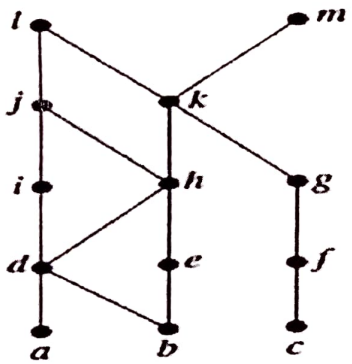
[10] (CO2/K3)

05. Use the method of generating function to solve the following recurrence relation.

$$Q_n = 2Q_{n-1} + Q_{n-2},\ n \geq 2,\ Q_0 = Q_1 = 2.$$ (Pell–Lucas Sequence)

[10] (CO3/K3)

06. For the partial order represented by the Hasse diagram as given below. Analyse the following
(a) Find the maximal elements (1 mark)
(b) Find the minimal elements (1 mark)
(c) Is there a greatest element? (1 mark)
(d) Is there a least element? (1 mark)
(e) Find all upper bounds of {a,b,c} (1 mark)
(f) Find the least upper bound of {a, b, c}, if it exists (1 mark)
(g) Find the lower bounds of {f,g,h} (1 mark)
(h) Find the greatest lower bounds of {f,g,h}, if it exists (1 mark)
(i) Find the lower bound of {l,m} (1 mark)
(j) Find the upper bound of {d} (1 mark)



[10] (CO4/K2)

07. a) Reduce the Boolean expression :$XY + X(Y + Z) + Y(Y + Z)$. (5 marks)
b) Draw a graph that has exactly two cut vertices and three cut edges. Justify your answer (5 marks)

[10] (CO4/K3)

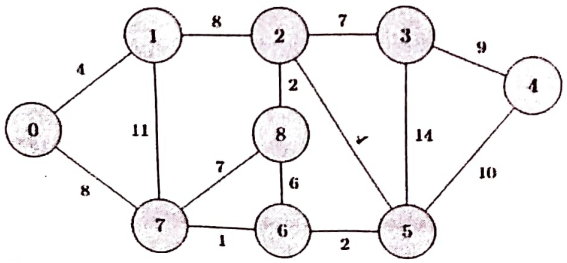
08. a) Examine whether the following two graphs G and G' associated with the following adjacency matrices are isomorphic (7 marks)

$$\begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \text{ and } \begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}.$$

b) Give an example of a graph that is an Euler graph but not Hamiltonian (3 marks)

[10] (CO5/K2)

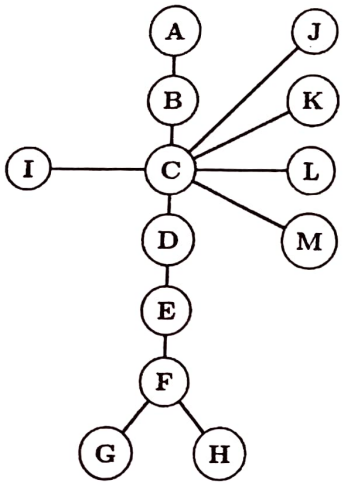
09. Using Prim's algorithm, find the minimum spanning tree of the following graph.



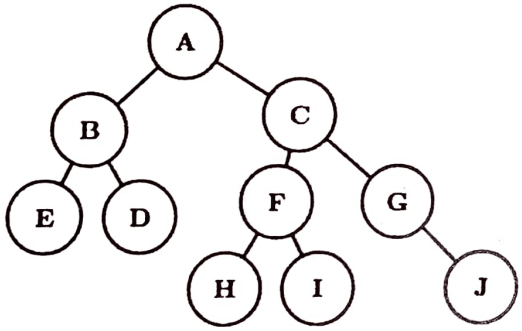
Hence, verify the minimum spanning tree with the Kruskal's algorithm.

[10] (CO5/K4)

10. a) For the following tree, (i) find the eccentricities of all the vertices (3 marks) (ii) find the diameter and centre (2 marks)



b) Find the in-order, pre-order, and post-order traversal of the following tree. (5 marks)

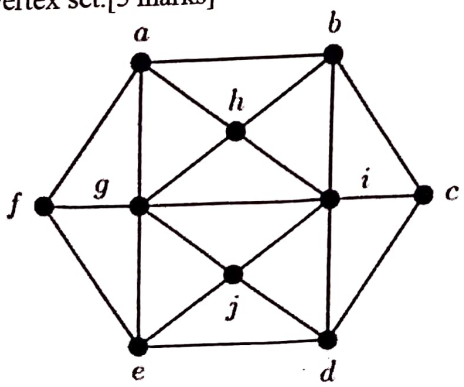


[10] (CO5/K3)

11. Seven employees are to be assigned to seven projects. Each employee can handle only certain projects based on their skills:
Arun:$P_1, P_2, P_3, P_4, P_5, P_6, P_7$; **Bina:**$P_1, P_2$; **Chetan:**$P_2$; **Divya:**$P_3, P_6$; **Esha:**$P_4, P_5$; **Farhan:**$P_4, P_6, P_7$; **Gopal:**$P_6, P_7$.
a) Draw a bipartite graph representing employees and projects with edges for valid assignments. (2 Marks)
b) Determine maximum matching, list all matched pairs and justify why it is maximum. (2 Marks)
c) Determine whether a perfect matching exists. Justify (2 Marks)
d) If employee Arun becomes unavailable, determine the maximum matching size in the new resulting graph and state whether it still admits a perfect matching.(2 Marks)
e) Find a minimum vertex cover and list its vertices. (2 Marks)

[10] (CO5/K4)

12. a) Determine the chromatic number of the given graph and obtain the corresponding chromatic partition of its vertex set.[5 marks]



b) Show that $k^3 - 4k^2 + k$ is not a chromatic polynomial for any graph. [5 marks]

[10] (CO5/K3)

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Reg. No

Final Assessment Test(FAT) - NOV/DEC 2025

Programme	B.Tech.	Semester	Fall Semester 2025-26
Course Code	BMAT205L	Faculty Name	Prof. Tharasi Dilleswar
Course Title	Discrete Mathematics and Graph Theory	Slot	C1+TC1+TCC1
		Class Nbr	CH2025260100879
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01. a) Establish the validity of the argument: "All integers are rational numbers. Some integers are a power of 2. Therefore, some rational numbers are a power of 2". **[5 marks]**
b) Consider the following statements:
(i) If he has dinner, then he does not drink tea.
(ii) He eats fruits only if he drinks tea.
(iii) He does not take juice unless he eats fruits.
Can we conclude that he takes juice whenever he has dinner? If not, what is the correct conclusion? **[5 marks]**
[10] (CO1/K3)

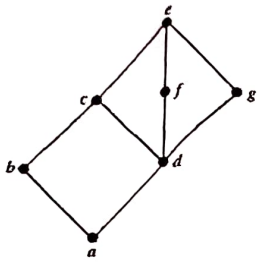
02. Consider the set $Q_8 = \{\pm 1, \pm i, \pm j, \pm k \mid i^2 = j^2 = k^2 = -1, ij = -ji = k, jk = -kj = i, ki = -ik = j\}$. By constructing the Cayley table, verify whether Q_8 is an abelian group with respect to multiplication.
[10] (CO2/K3)

03. a) Prove or disprove that the union of two subgroups of a group G is a subgroup. **[3 marks]**
b) Let $G = \mathbb{Z}$, be a group of set of integers under addition and $H = 3\mathbb{Z}$, be a subgroup of G. Find the cosets of H in G. **[4 marks]**
c) Determine whether the mapping $f : \{\mathbb{Z}_4, +_4\} \longrightarrow \{\mathbb{Z}_{10}, +_{10}\}$ defined by $f(x) = 2x$ is a group homomorphism. Justify your answer. **[3 marks]**
[10] (CO2/K3)

04. Solve the recurrence relation $a_n - 3a_{n-1} + 2a_{n-2} = n, \quad n \geq 2$, given that $a_0 = 1, a_1 = -1$, using generating function.
[10] (CO3/K2)

05. a) Without constructing a truth table, simplify $\phi = [(p \to q) \wedge (q \to r)] \vee [(q \to p) \wedge (r \to q)]$ and determine whether it is a tautology, contradiction, or contingency. **[5 marks]**
b) Among 500 students in a college, 65 learn French, 45 learn German, and 42 learn Russian. If 20 learn both French and German, 25 learn both French and Russian, 15 learn both German and Russian, and 8 learn all three, find the number of students who study none of these languages. **[5 marks]**
[10] (CO3/K3)

06. a) Draw the Hasse diagram for the poset A={2,3,6,12,24,36,72} under the relation of divisibility, and check whether the poset is a lattice. **[5 marks]**
b) Check whether the given lattice is distributive and/or complemented. **[5 marks]**



07. a) Find the sum of products of $((xy)'z)'((x'+z)(y'+z'))'$ without using truth table. **[5 marks]** **[10] (CO4/K4)**

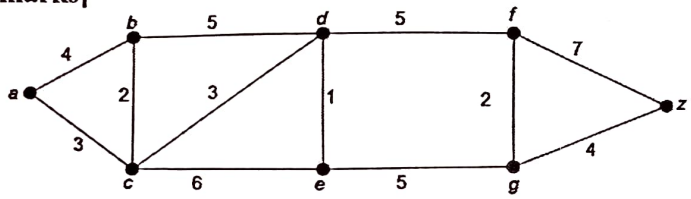
b) Draw a graph having the adjacency matrix

$$X = \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

with respect to the ordering of vertices a, b, c, d, e and find the number of walks of length 2 between every pair of vertices. **[5 marks]**

[10] (CO4/K2)

08. a) Use Dijkstra's algorithm to find the shortest path between the vertices a and z in the below weighted graph. **[7 marks]**



b) Find 3 non-isomorphic graphs on 5 vertices. **[3 marks]**

[10] (CO5/K3)

09. A utility company needs to lay down electrical cables to connect five different towns: Town A, Town B, Town C, Town D, and Town E. The company has multiple options for connecting each pair of towns, each with different costs (in thousands of dollars).

The costs (in thousands of dollars) for connecting the pairs of towns are as follows:

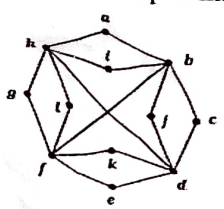
Edge	Cost (in $1000)
A – B	12
A – C	8
A – D	10
B – C	9
B – D	6
B – E	7
C – D	5
C – E	11
D – E	4

Find the minimum cost required to connect all the towns using the minimum spanning tree (MST) approach, and list which connections (edges) should be used.

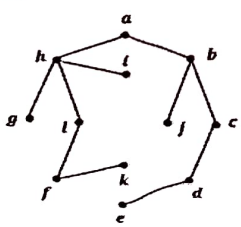
[10] (CO5/K3)

10. For the given graph G, identify at least five fundamental circuits and fundamental cut-sets formed with reference to its spanning tree T.

G

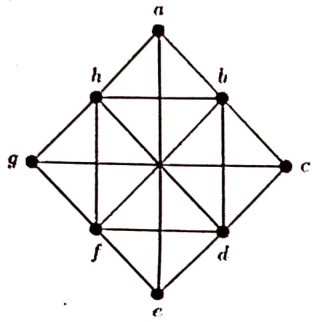


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[10] (CO5/K4)

11. Given the graph below, answer the following:
a) Find a maximum matching in the graph and check whether it is a perfect matching. **[5 marks]**
b) Find a minimum vertex cover for the graph. **[5 marks]**



[10] (CO5/K3)

12. Suppose that in one particular semester, there are students taking each of the following combinations of courses.
* Mathematics, English, Biology, Chemistry
* Mathematics, English, Computer Science, Geography
* Biology, Psychology, Geography, Spanish
* Biology, Computer Science, History, French
* English, Psychology, History, Computer Science
* Psychology, Chemistry, Computer Science, French
* Psychology, Geography, History, Spanish.

a) What is the minimum number of examination periods required for exams in the ten courses specified so that students taking any of the given combinations of courses have no conflicts? **[5 marks]**
b) Find a possible schedule which uses this minimum number of periods. **[2 marks]**
c) Draw the bipartite graph for the given student-course combinations and explain how this representation helps in identifying course conflicts for exam scheduling. **[3 marks]**

[10] (CO5/K4)

BL-Bloom's Taxonomy Levels - (K1-Remembering,K2-Understanding,K3-Applying,K4-Analysing,K5-Evaluating,K6-Creating)

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